Examiner only

Answer **all** questions in the spaces provided.

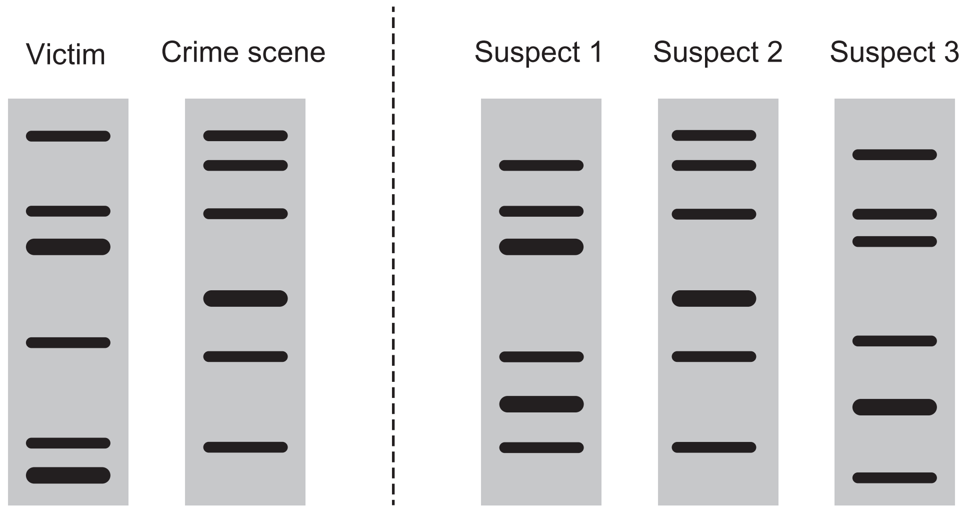
1. Criminals will try to hide their activities but forensic scientists, using biological and chemical testing, are able to find evidence that reveals what happened.

(a) DNA profiling is commonly used to identify criminals.

(i) State where DNA is found in the cell. [1]

....

(ii) There were three suspects in a crime. Suspect 1 did not have an alibi but suspects 2 and 3 did. Initially it was thought that suspect 1 was the criminal. DNA samples were collected from the crime scene.



Compare the DNA samples above to explain whether having an alibi or not is sufficient to decide on guilt. [2]

....

....

....

(iii) State **one other** use of DNA profiling. [1]

....

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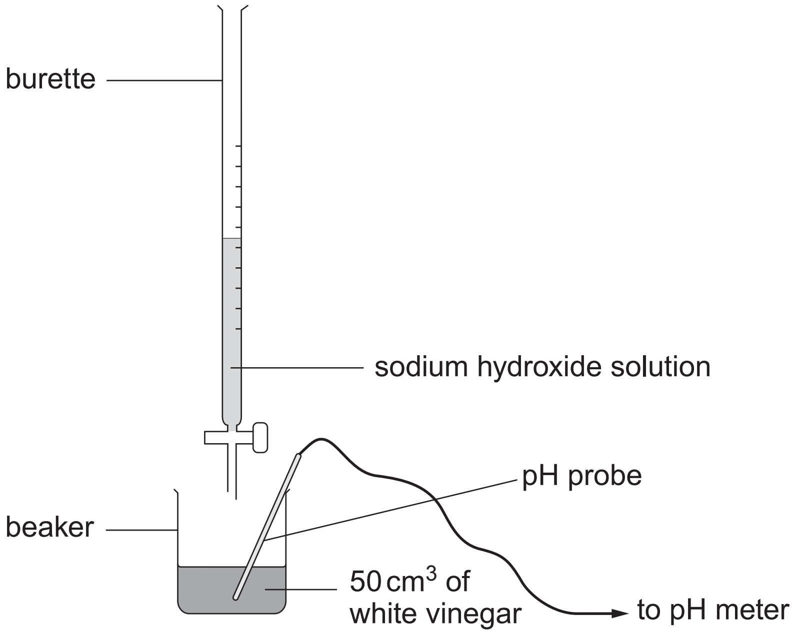
 (3445UC0-1) **Turn over.**

(b) It is suspected that supplies of vinegar have been tampered with by adding a strong acid. White vinegar usually contains ethanoic acid at a concentration of 0.01 mol/dm^3. Samples of the vinegar are taken and tested by titrating against 0.5 mol/dm^3 sodium hydroxide solution using the apparatus below.



A pH meter is used to monitor the pH of the solution in the flask. The results are shown in the table below.

Volume of sodium hydroxide solution (cm^3)	pH reading
0	1.5
5	1.6
10	1.7
15	1.9
20	2.2
25	3.0
30	13.7
35	14.0
40	14.0

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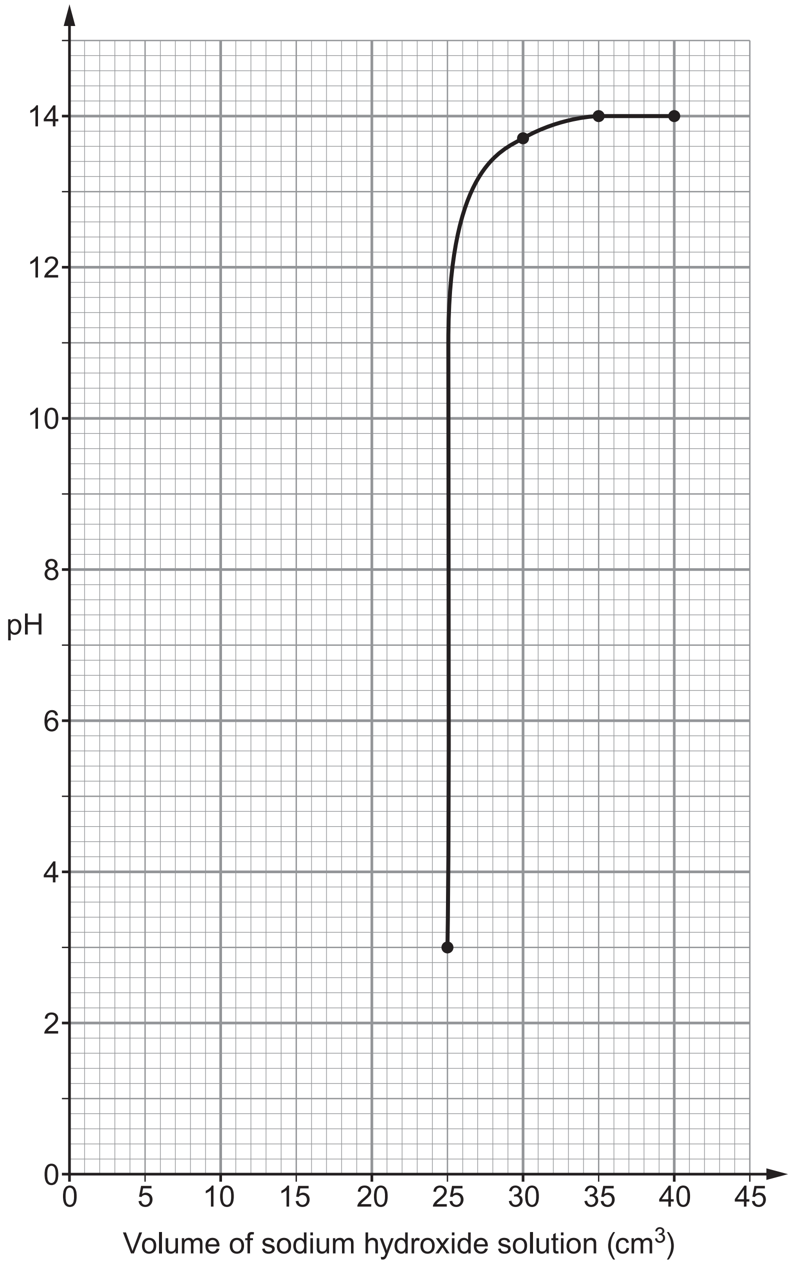
(i) Use the data in the table to complete the graph on the grid below. [3]



(ii) Use the graph to find the volume of sodium hydroxide solution required to neutralise the white vinegar. [1]

volume = ... cm^3

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(iii) Use the information on pages 4 and 5 and the equation

$$\text{concentration of white vinegar} = \frac{\text{concentration of sodium hydroxide} \times \text{volume of sodium hydroxide}}{\text{volume of white vinegar}}$$

to determine whether or not the samples had been tampered with. [3]

....

....

(iv) Complete the word equation for the reaction between an acid and a base. [2]

acid + base → ....

(v) Explain why taking multiple readings will increase the accuracy of the result. [2]

....

....

....

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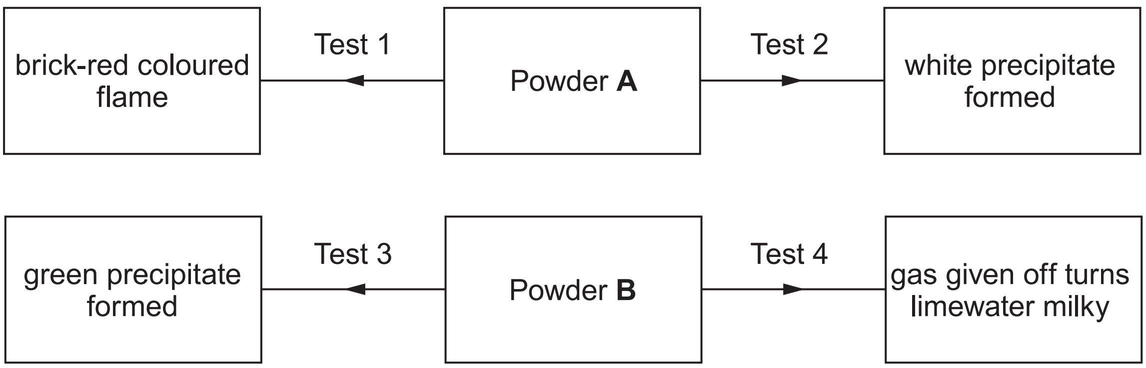
(c) Powders **A** and **B** are collected at a crime scene. The following tests are used to identify them.

Test	Action
1	Flame test
2	Add silver nitrate solution
3	Add sodium hydroxide solution
4	Add dilute hydrochloric acid and bubble gas through limewater

The results are given below.



Identify powders **A** and **B**. [4]

Powder **A** ....

Powder **B** ....

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